

UnlockMe: Social Interactions when Co-located in Online Activities

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ABSTRACT

All the activities that we do online, by either preference or obligation, deprive us of social interactions especially the impromptu ones. We present UnlockMe, a concept that aims at preserving the social link, the disposition to come across other people serendipitously when engaged in online activities such as purchasing goods, working from home or when viewing media and entertainment. Our concept relies on virtual co-location detection (both synchronous and asynchronous), to allow users to engage with other people with whom they would have been likely to interact when doing the same activities offline in the physical world. We developed and illustrated UnlockMe with six scenarios and low-fidelity prototyping to test it with 10 participants who were isolated due to the coronavirus disease (COVID-19) pandemic. Our findings reveal multimedia recommendations from close social connections to be the best scenario for UnlockMe followed by online shopping and connecting with the local community.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Collaborative and social computing theory, concepts and paradigms; Social networks; Human computer interaction (HCI); Interaction paradigms; Collaborative interaction.

KEYWORDS

Social Connections, Virtual Co-location, Serendipitous Messaging, Network Segregation, Recommendations, Collaborative Decision Making

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1 INTRODUCTION AND RELATED WORK

People who are isolated and confined to their homes due to illness, temporary or permanent physical challenges, old age, remote work or a global pandemic rapidly lose basic social interaction. During these times, remotely interacting with other people is an elementary need of any human and key to one's mental health [31]. As observed during the current coronavirus disease (COVID-19) pandemic [39], the use of multimedia communication tools (text messaging, voice or video calling or conference meeting) and social media platforms rose to fill the need of connecting with other people during physical isolation [16] [38]. Although social media helped people to connect with outside world during this time, it has also been strongly associated to mental health problems [9] [34]. The social media which is on the rise since early 2000's [24] with increasing usage during the pandemic, may have at least partially failed in fulfilling our need for meaningful connections [26]. Indeed some prior research finds a correlation between social media exposure and an increase of social isolation, depression, anxiety, upward social comparison, jealousy and loneliness [25] [26] [36] [40].

The importance of informal interactions between colleagues aiming to build and preserve social connections beyond information exchanges, theorised as "outerraction" [21] has been of significant focus in the CSCW and HCI communities. This outerraction has migrated to social media in recent years putting people in awkward positions where they try to restrict their colleagues from prying into their personal lives [35]. Although Twitter [32] has become a successful micro-blogging support for outerraction at work [41], like other social media and instant messengers [23] it does not offer segregation of social groups. This flattening of multiple contacts to a single group pushes users to manually distribute and restrict different contacts to different applications and researchers identify the need to support different communications spaces with the same tool [23] [35].

Before the advent of social media, virtual co-location concepts like WebWho [17] and WebRogue [29] aimed to provide awareness to virtually co-located people, to facilitate serendipitous collaboration when they are involved in activities such as attending a lecture together, group surfing and waiting in line. These first investigations were based on considering the Web to be a social place where people perform activities and therefore aim at making their presence and activity visible and exploitable by other social inhabitants of the same virtual location. These initial efforts lacked the abilities to categorize online activities (beside exact virtual location), to partition networks and to switch from synchronous to asynchronous communication. These first investigations have been followed by the raise of fully synchronous and asynchronous dedicated spaces: social media lets us know who is connected at a given time, while media spaces inform us what other people with similar taste have

been reading, watching and buying inside the same channel. Our investigation is looking at the possibility of an awareness support that is at the same time more pervasive, cross channelled, and with a notion of different social circles.

In the rest of this paper we introduce our concept, present our preliminary remote user study, report and discuss our findings and conclude with reflections on our limitations and future work.

2 UNLOCK ME: THE CONCEPT

UnlockMe is a concept to virtually co-locate users in their online activities and provide them with a channel for synchronous or asynchronous communication across the web, for a broad variety of activities that could be performed on the web such as entertainment, work and shopping. Our concept differentiates from social media such as Facebook [8] and Instagram [14], by being centred on activity triggered social exchange, contrary to providing multimedia content and advertisements for absentminded scrolling [20]. We also differentiate from messaging apps such as WhatsApp [37] as the user does not need to move away from their current web activity to another messaging app and they can meet people serendipitously, without setting up dedicated chat groups for each topic or each contact network.

This choice resulted in the design of the concept as a meta cross service functionality which moves away from the attempt to maximise the time spent by a user in a particular virtual space (website or application) and from the need to move to a dedicated synchronous communication channels (chat apps). The key objective is to provide opportunities for social exchanges, triggered by mutually interesting activities. In this sense, the service is aiming at providing both the trigger and support for communication, across the variety of activities a person may perform online. UnlockMe is a remote synchronous, serial and asynchronous groupware [7] that connects users to their various real-life social networks (friends, family, colleagues, neighbourhood) during various online activities, allowing them to casually interact, using more or less similar activities as triggers for the exchanges. To obtain this, it relies on the concepts of networks segregation and virtual co-location.

Networks segregation (or “Audience segregation”) is the partitioning of different audiences and the compartmentalisation of social spheres. In social media, it was proven useful to protect users’ privacy and improve quality of interactions [2]. Virtual co-location is the idea of detecting when the user is close, by their online activity, to another user. Even if it allows to group users online, our definition is quite different to the one of [6] where they had built a tool to manually create such co-location. This concept of co-localisation works at different scales, from a broad-level of “entertainment”, “purchase” or “work” to a narrow-level of a particular web link such as certain shopping item or a video. In order to detect these activities at different levels, the tool needs to analyse the current content of the user’s screen. We can think of several techniques to classify online activity including URL parsing [10], image classification of the browser or desktop app screen [11], or even a hybrid combination of both.

For the initial conceptualisation we choose the form of a browser extension for UnlockMe to co-locate users performing online activities in the same websites using a web browser. However, our

concept could also be implemented at the level of the operating system to cover the online activities done with dedicated apps on desktop and mobile devices. We also chose text messaging for the communication, but other forms such as audio, video and screen-sharing would be equally possible with our concept.

3 REMOTE USER STUDY

To test the concept of UnlockMe and collect early end-user feedback, we ran an experience prototyping [3] remote user study. 10 participants, who were quarantined and working from home during the coronavirus disease (COVID-19) pandemic [39] participated in the study over individual Zoom meetings [42]. Two researchers conducted the study. We started the session by discussing their current day-to-day life in quarantine, leading them into the topic of social interactions. As a next step, we demoed a low-fidelity clickable browser plug-in of UnlockMe designed in Sketch [27] and hosted with InVision [15]. Using the prototype and the primary interactions such as awareness of co-located online presence, personal messaging and group broadcast (as shown in Figure 1) we explained our concept to the participants. We then discussed six scenarios (described below) for leisure activity related social interaction aiming at outerraction [21] rather than work and productivity, for using UnlockMe on top of the corresponding websites as shown in Figure 2

3.1 Awareness of Colleagues Browsing Wiki

Awareness is a key concept of collaborative communication [17] [19] which simulates togetherness, informs of availability and can lead to synchronous communication. In the first scenario, participants were asked to imagine visiting Confluence [5] (a web-based corporate wiki). As UnlockMe is currently activated they become aware of other colleagues who are doing the same activity at the same time. Without engaging in explicit interactions, they merely gain awareness about the presence of others. We probed the participants on how they felt about being aware of their colleagues’ synchronous presence at the same website.

3.2 Synchronous Connection to Chat in E-mail

Colleagues who are checking e-mails using a web browser at the same time are aware of other colleagues present online and broadcast messages to connect for a virtual break-time in the second scenario. Here we noted the participants’ preferences for interacting with colleagues or other contacts for an informal discussion and under what circumstances they would be most willing to communicate synchronously with UnlockMe.

3.3 Asynchronous Connection to Receive Recommendations in Netflix

In the third asynchronous scenario (same website and different time) participants were asked to imagine visiting Netflix website [22] and see a message recommending content from their friend/family member/colleague who is currently offline. We investigated whether the participants would appreciate this recommendation and be likely to leave recommendations for their social connections with UnlockMe.

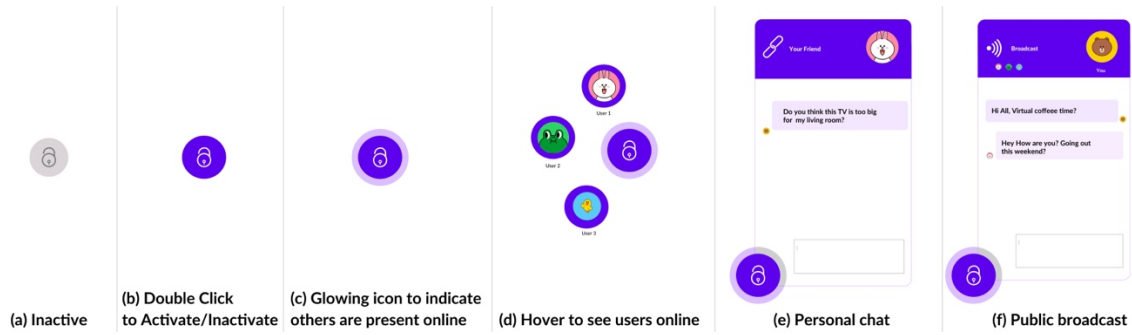


Figure 1: Interactions chosen for the initial conceptualisation of UnlockMe.

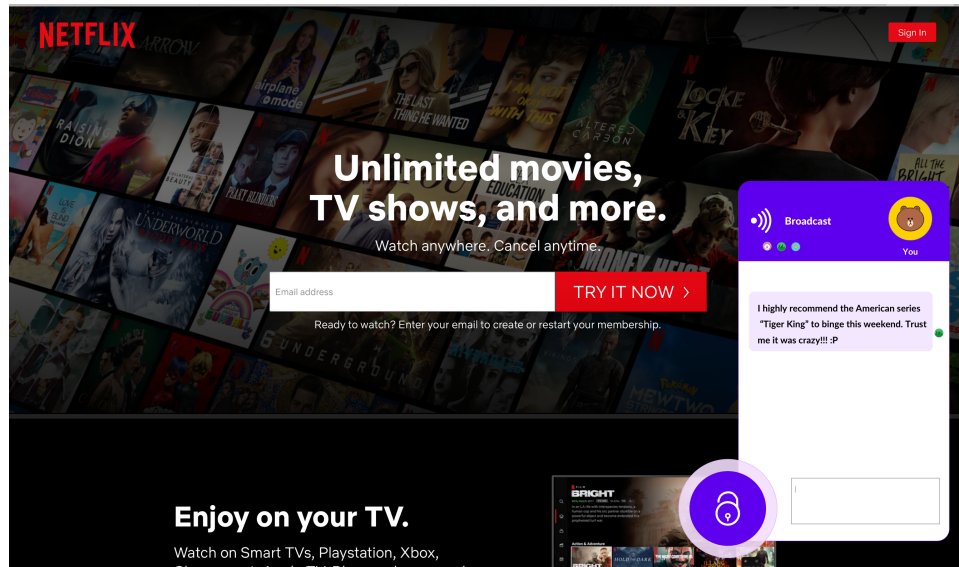


Figure 2: Scenario of user receiving recommendations from their friend/family in Netflix via UnlockMe.

3.4 Debating News

On a blog or a news website, the participant may read comments that were broadcasted by contacts in his/her social circle in the fourth scenario. We asked whether they would use such feature to watch or participate in the debate of a controversial article with a social contact who has read and liked it or broadcasted a comment to a given personal network to trigger a discussion. We emphasised on how UnlockMe's feature was different from leaving a public review on news website or sharing links on messenger apps.

3.5 Synchronous or Asynchronous Shopping

The tool could also allow the participant to check with a friend or family whether the television model which they are considering to purchase will be suitable for the living room. We probed the participants about their interest in the capability to bring this contact directly to the merchant's website to discuss products specifications or performances both synchronously and asynchronously in the fifth scenario.

3.6 Connecting with the Local Community

In the last scenario, we moved outside the pre-existing social circle of the participant. If they were buying groceries (for home delivery) from a local provider and we asked how UnlockMe could ignite the kind of discussion one may have when visiting a local shop. Here, we inquired them on accepting recommendations from strangers in their locality, meeting local friends/acquaintances on the web and communicating with the local shop/restaurant owners.

For each of the above discussed scenarios we recorded the participants' perceived future use of UnlockMe (as shown in Table 1). We concluded the session by discussing and capturing subjective measures of the usefulness and usability of UnlockMe. We also probed the participants on the situations and people with whom they are likely to use our concept. The sessions lasted about 45 minutes each and the Zoom meetings were recorded and transcribed. The data were open coded and thematically analysed [4] in separation by the two researchers. A consensus was reached on the resulting themes after three meetings. Participants received few goodies such as pens and tote bags as a remuneration.

Table 1: Participant profiles and colour coded (Purple = Definitely, Light Purple = Maybe and Black = Never) perceived use of UnlockMe in the scenarios discussed.

P.No.	Sex	Age	Residence	Occupation	Confluence	E-mail	Netflix	News	Shopping	Locality
P1	M	35-39	Family	Sr. Engineer	Purple	Black	Purple	Purple	Purple	Purple
P2	M	30-34	Single	Jr. Engineer	Purple	Black	Purple	Purple	Purple	Purple
P3	M	50-54	Family	Team Manager	Light Purple	Purple	Purple	Black	Purple	Purple
P4	M	30-34	Couple	Jr. Scientist	Light Purple	Black	Purple	Light Purple	Light Purple	Purple
P5	F	45-49	Couple	Sr. Scientist	Light Purple	Black	Purple	Black	Purple	Light Purple
P6	F	20-24	Couple	Intern	Light Purple	Black	Purple	Light Purple	Purple	Purple
P7	M	20-24	Single	Intern	Purple	Black	Purple	Black	Purple	Light Purple
P8	F	50-54	Single	Executive Assistant	Black	Black	Purple	Light Purple	Black	Black
P9	F	25-29	Family	HR Assistant	Purple	Black	Purple	Light Purple	Purple	Light Purple
P10	M	25-29	Single	PhD Student	Purple	Black	Purple	Black	Purple	Purple

4 FINDINGS AND DISCUSSION

Most participants (8/10) found UnlockMe to have the advantage over sharing long web-links and screenshots for the scenarios discussed, which leads them to switch between different devices and applications. All participants (10/10) found UnlockMe to be easy-to-use and useful for them in situations of isolation for one or more scenarios. They were excited to receive personal and serendipitous messages and felt less isolated as reflected in the quotes below:

“I like the fact that this is not just a chat or social network, it is in general and it can be synchronous or asynchronous messaging.” (P2)

“I think this would make me feel less lonely in confinement. We can see that people are there and we can choose to reach out or not instead of setting numerous formal meetings” (P6)

“Originality of the idea, I don’t think something like this exists. I like the fact that this could lead to serendipitous meeting that otherwise would never happen” (P9)

Most participants (9/10) liked being aware of other colleagues who are virtually co-located with them in Confluence at the same time. However, having the possibility to interact with colleagues while checking e-mail was heavily disliked by most (9/10). Participants suggested they would be interested in interacting with their colleagues “while waiting for things to upload or download in drive” (P2), “while searching to solve a problem in Stack Exchange” (P9) or “during long Zoom calls” (P10). None of them (0/10) said they would definitely engage in debating news with their social circle. We discuss the major themes resulting from the analysis of our study in this section.

4.1 UnlockMe for Reliable Recommendations and Collaborative Decision Making

We found the Netflix scenario, for exchanging streaming content recommendations with trusted social contacts to be the best-liked by all our participants (10/10) as expressed by P3 and P9 here:

“We do that on WhatsApp now, I’ve seen that on Netflix” but having that in a widget and sharing that with my friends is better. I think it

accelerates things vs. taking a screenshot of something, sharing with another whatsoever medium, everything will be combined here and it is easy to use.” (P3)

“I think I would enjoy it. To know that someone I know had left a recommendation in Netflix and if I know this person to have good taste in movies!” (P9)

The synchronous or asynchronous shopping scenario was also appreciated by most participants (9/10). The example of shopping cued our participants to discuss more perceived collaborative decision-making use cases for UnlockMe in their lives such as booking tickets/hotels, planning parties and preparing outings as discussed by P9 and P10 below:

“Actually, this just gave me an idea about where I might use it. For example, if I plan some holidays with friends and we are looking for a good place in Airbnb, this could be very useful, I think it is tedious to copy and paste so many links and ask what do you think about this one. There it would be very practical to have such a tool.” (P9)

“Maybe I will share videos in YouTube with my friends. I feel good about this, it is nice. I prefer asynchronous communication more for YouTube. For shopping I would like to do it together at the same time. A lot with my family and sometimes with my friends. It depends with my family though, I might phone them too. With friends this is useful if we are buying something for a birthday.” (P10)

Our findings suggest that receiving recommendations from personal social circles for online entertainment and shopping is greatly preferred to scrolling recommendations from recommender systems. UnlockMe would foster close social bonding with human-human recommendation, with smart sharing mechanisms such as automatic contact and recommendation suggestions.

4.2 UnlockMe for Connecting with the Local Community

Connecting with the local community also proved to be an interesting scenario for most participants (9/10). They envisioned UnlockMe to be useful for interacting with local people for

information or random chit-chats in the websites of events, educational institutions, local shops/restaurants and governmental/non-governmental organisations as in their comments below:

"Maybe for recommendation on groceries I could ask question. I could also connect to a person from the store, even if I don't know if I'd trust him/her. For Do-it-yourself, I wouldn't mind technical recommendation or Q&A from the shop owner." (P3)

"I like the idea of recommending things to people in same grocery shop, buyers have something to say, a hygiene problem for instance. Not only I missed it during quarantine, but I think this local information sharing is always missing but important: where is the best place to buy this kind of item . . . this supermarket has this product cheaper than where I go. Or give a confinement specific information, the hydro-alcoholic gel is available there, hurry to buy it: very useful information." (P4)

"I always try to see local food recommendations from colleagues or the internet. So, for me reviews and recommendations are very important. Maybe restaurants from local community and maybe shops. If I'm new to a place then probably I'll look for sightseeing places as well, quiet and peaceful and non-polluted areas to exercise" (P6)

With local community focused research and applications being identified as a re-emerging field of interest in recent years [18], in the very peculiar situation of a pandemic induced lockdown, connection with local community is indeed very important [1] [12]. The use of such tool could help maintaining a social link and staying aware of the resources nearby (information, goods and services) in the future. Also, local business owners could benefit from smooth interactions with their customers using UnlockMe.

4.3 User Group Segregation and Privacy Concerns

All participants (10/10) wanted some kind of user group segregation when using the system. They wanted to have different channels of UnlockMe for personal and professional contacts.

"Maybe privacy is a problem. I would like to have a difference between professional and personal communities. I would like to have separation between colleagues and friends so my colleagues don't see the same things as my friends and family" (P2)

While our participants felt secure about revealing their identity to their pre-established personal and professional contacts, with public contacts in the locality scenario where it was possible to talk to a stranger or shopkeeper they raised concerns as expressed by P3 and P9 here:

"I would appreciate to quickly separate communities and, of course, the ability to say 'yes, I am okay with that or I'm not okay' choose just by clicking 'I want to share'. Allow to be sort of 'Incognito' somewhere." (P3)

"I think I will be suspicious that it might be a bot by the company or someone paid by the company to say a product great. However, if its someone with their Facebook ID I think I might trust. But I'm not sure if I want to broadcast my real name. In real life you know the person you are talking to is the person you are talking to but online you never know." (P9)

Whether it is in real-life or online, one would not like to share the same things to different kinds of contacts [23]. A very important

aspect of UnlockMe that resonated with the users was the ability to share (communicate status, broadcast comments, receive notifications) with a selected population. Our initial concept allows user's contacts to be segregated in contact categories, either standard (friends, family, colleagues, local community) or user defined. Further, communications about a given activity could be proposed to the appropriate category of contacts. While the user can always classify her/his contacts manually, after some interactions, this segregation could also be automatically proposed by the tool. In the same way, websites could also be classified as appropriate for a given circle (friends, family, colleagues or local community) but the user could effortlessly change this circle to engage with another.

4.4 UnlockMe as a Private Media for the Young

Most of the participants above 30 years of age (5/6) saw UnlockMe as yet another tool for communication which they might be willing to try for the selective purposes that interests them as anticipated by P1 and P5 below:

"I am so busy at home and I don't feel the need of having an additional way of interacting with the people. I have the kids, the school, the work and the home, I never feel lonely. Could be the work environment, I feel a bit disconnected. We have the chat, but the added value for this (UnlockMe) is that you know that the connected people are also available." (P1)

"I'm not so convinced as I'm old and this is for younger people. The thing I have in my mind is this is 'yet another thing', but I could try with close people" (P5)

On the contrary, all younger participants (4/4) perceived UnlockMe to become a contextualised private communication channel in comparison to social media applications such as Instagram [14] and Facebook [8] which they were currently using, and they liked this change as expressed by P6 and P7 here:

"If my friends are using UnlockMe, I don't have to post on Facebook or Instagram stories for everyone to see. Also, for shopping I ask them right away from the buying page itself, this might be easier for me." (P6)

"For example, in Facebook usually the post is public and everyone can still see what I'm saying, maybe from the perspective of my friends its ok but to others it can be offensive" (P7)

This finding again emphasises the need for network segregation in social tools. It also denotes the willingness of the younger generation to move from content based social media and try new tools for social connections like UnlockMe.

5 LIMITATIONS AND FUTURE WORK

While the findings and discussions from our user study are essential material for the continuation of our concept exploration, we acknowledge that they are preliminary. Indeed, we recruited a rather small user sample size, with possible bias in recruitment (location, income level, technology savviness). We also presented a low-fidelity prototype and scenarios that were showing early stage ideas, leaving room for discussion, interpretation and ideation with the participants.

To move forward with our research, our future steps would be to develop a high-fidelity prototype, based on the preliminary

uncovering and perform longitudinal testing. We would build it for receiving reliable recommendations, collaborative decision making and connecting with the local community, supported by featuring strong audience segregation and co-location detection mechanisms. We would also test features that were hinted by our participants such as importance of a mobile version, time-limit for messages/recommendations and bookmarks.

To ensure that a concept for social connectivity does not become a commercial attention trap for a user in a particular virtual space and sustain genuine social interactions, we need to avoid the current ad-based business models [13] [30] [33]. In our concept, a suitable commercial model would be built on the value UnlockMe conversations bring to the providers of the online activities upon which the conversation occurs (for examples: news, media or e-commerce).

6 CONCLUSION

In this paper we presented and tested the preliminary concept of UnlockMe, for social interactions when co-located in online activities. In light of the needs and habits expressed by our study participants and the analysis of other studies situated in the current coronavirus disease (COVID-19) pandemic [1] [9] [31] [38], UnlockMe presents outerraction [21] opportunities overcoming some flaws and threats uncovered with mainstream social medias. While we do not claim that it can be a perfect substitute for physical social interaction, we believe that there is a strong interest for social interactions situated alongside online activities. All of our participants stated that they would like to work remotely for two or three days a week in the post-pandemic world, as working from home is here to stay [28] and this calls for more research on fostering social bonds. With a design that encourages genuine social interactions and provides networking suggestions with caution and transparency, we aim to further test a refined and more advanced version of UnlockMe.

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